

Emergency rooms keep their promises to the patients who need them **least**.

Vani Sahjwani · University of California, San Diego · SDUTC 2026

5.5%

of the **sickest** patients are seen within the clinical target. The goal is 98%.

95%

of the **least urgent** patients are seen within target. Their goal is only 80%.

11%

of a patient's triage score is explained by **which hospital** they walked into.

1 THE SETUP

What triage promises, and what we measured it against

The one rule triage has to follow

Every emergency room sorts patients into five urgency levels at the door. Level 1 is a cardiac arrest. Level 5 is a sore throat. Each level carries a published maximum wait, from **0 minutes** at level 1 to **120 minutes** at level 5.

The rule is simple: sicker patients should wait less. Hospitals audit their average wait times, but almost nobody checks performance **level by level** against the clinical standard. We did that nationally.

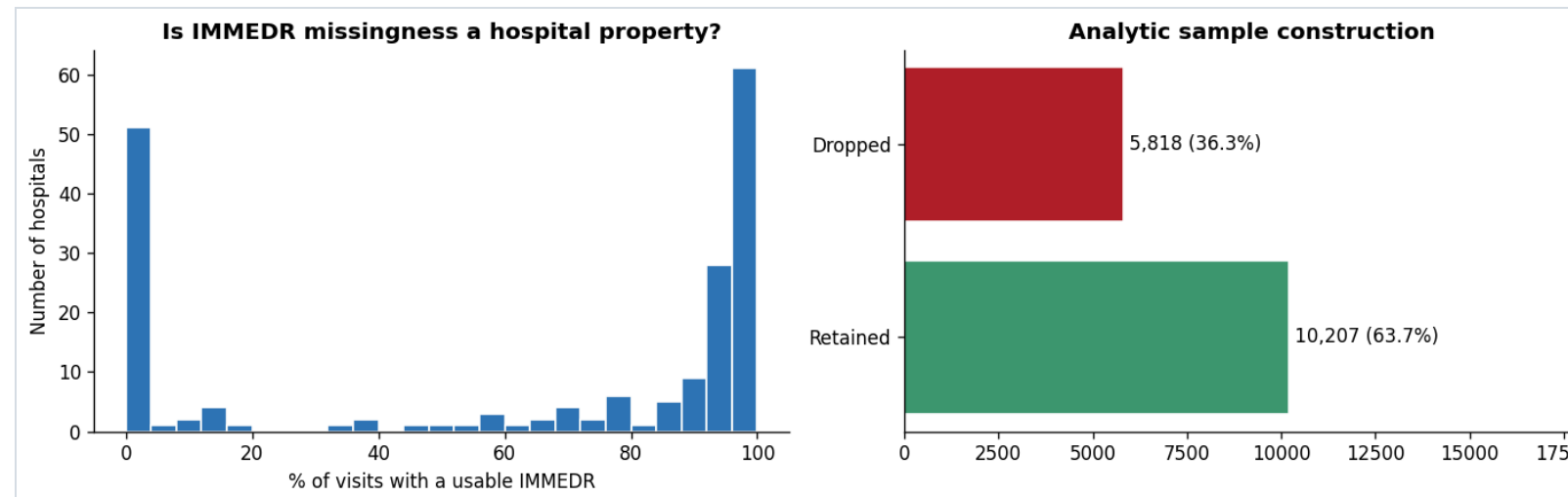
THIS STARTED AS AN AI PROJECT

The original plan was a model that predicts urgency at intake. We shut it down. Section 6 shows the number that killed it, and why that failure is what uncovered the real finding.

The data, and what it cannot tell us

NHAMCS is the CDC's national survey of emergency visits. 2022 is its final year. It is a weighted probability sample, so each record stands in for thousands of real visits: **16,025 records** become **10,207 usable** ones representing **101.5 million** encounters.

Using 2022 matters twice over. It is the last year this survey exists, so it is the final national picture of ED triage before surveillance changes. It also carries higher sampling error than earlier years, which is why every threshold in this project is set conservatively.

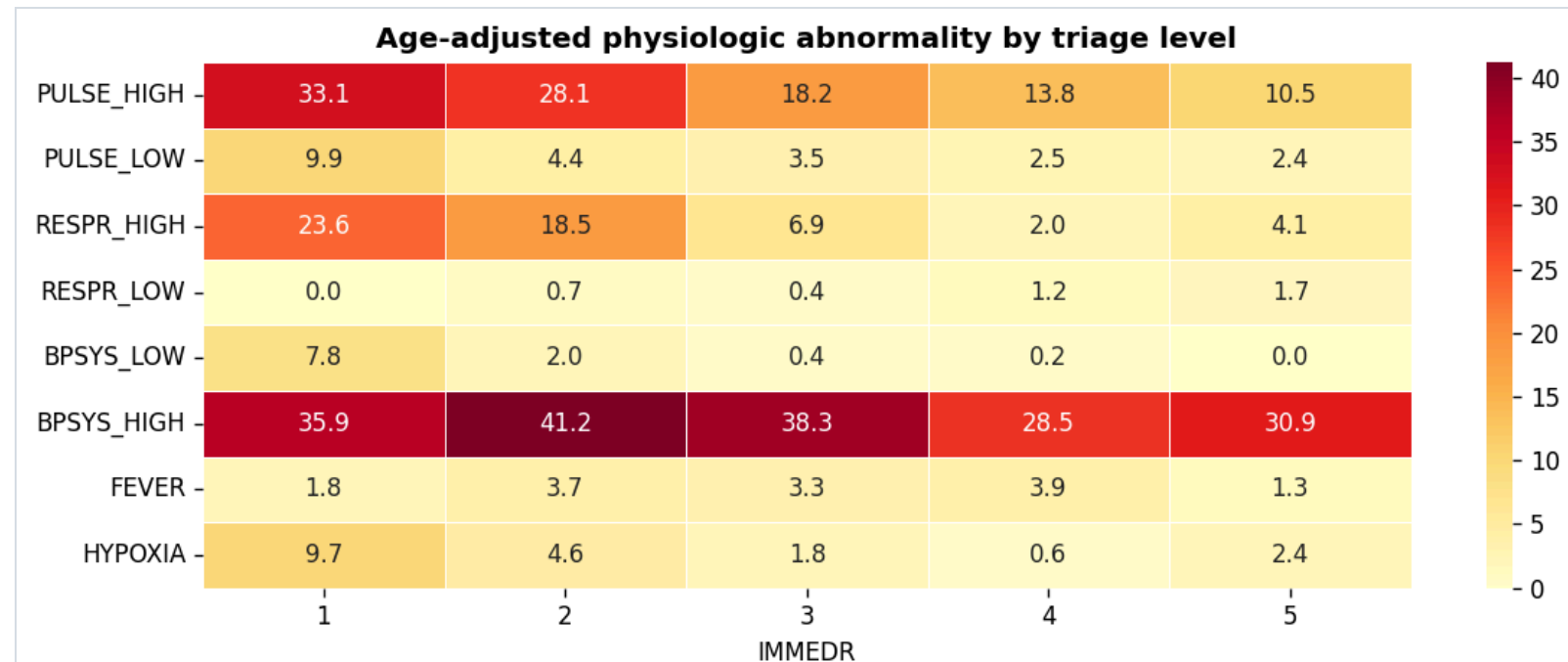


36% of records carry no triage level, and the gap clusters by hospital. 41 EDs never record it at all. We state this up front: our results describe hospitals that document triage, not all hospitals.

Four corrections most analyses skip

Each one changed a conclusion. Leaving any of them out produces a confident wrong answer.

- Survey weighting.** Because records carry unequal weight, 10,207 rows behave like only 5,266 independent ones. Skip this and every result looks twice as certain as it is.
- Hospital clustering.** Patients inside one ED are not independent of each other. Every interval here resamples **whole hospitals**, not rows (Rao and Wu cluster bootstrap, 1,000 draws).
- Age-adjusted vital signs.** A pulse of 120 is normal in a toddler and alarming at 74. Raw vitals confound every comparison with who happens to be in the room.
- Multiple testing.** Dozens of tests were run, so we test at $\alpha = 0.01$ and control the false discovery rate.



Age adjustment turns noise into signal. Raw pulse is flat across all five levels and looks useless. Once adjusted for age, abnormality falls cleanly from level 1 to level 5.

WHAT DID NOT WORK, REPORTED ANYWAY

Crowding test: load proxies were uncorrelated and every slope was null. **Discordance detector:** collapsed at $n=139$ and is reported as a failure, not a finding. **Censoring:** the CDC dropped the left-without-being-seen fields, so only a proxy was available. **Disparity by race, sex and payer:** no significant effect after correction.

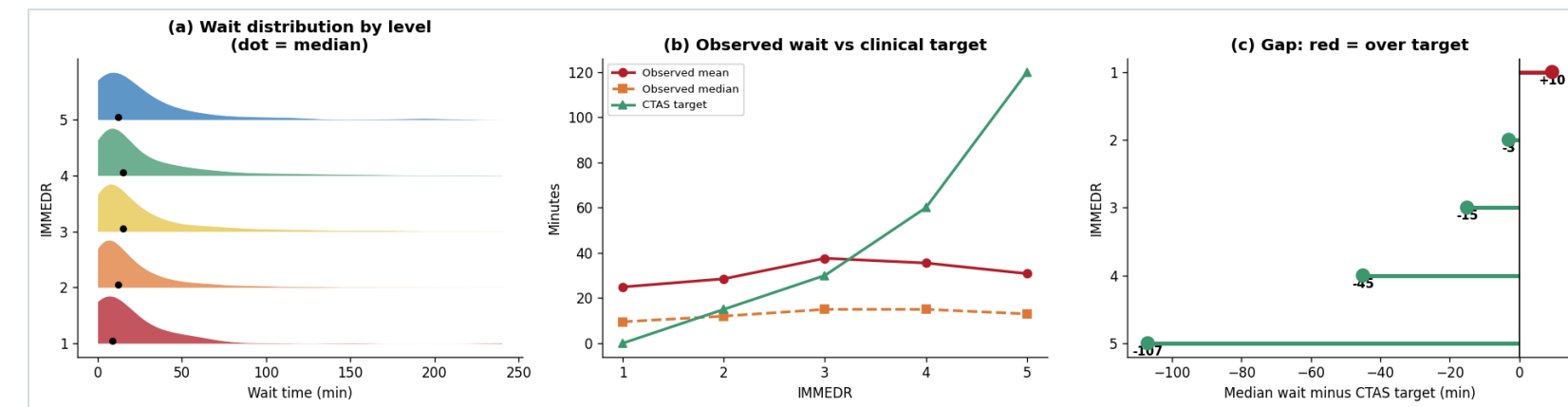
2 WHAT WE FOUND

Flat service measured against a steep standard

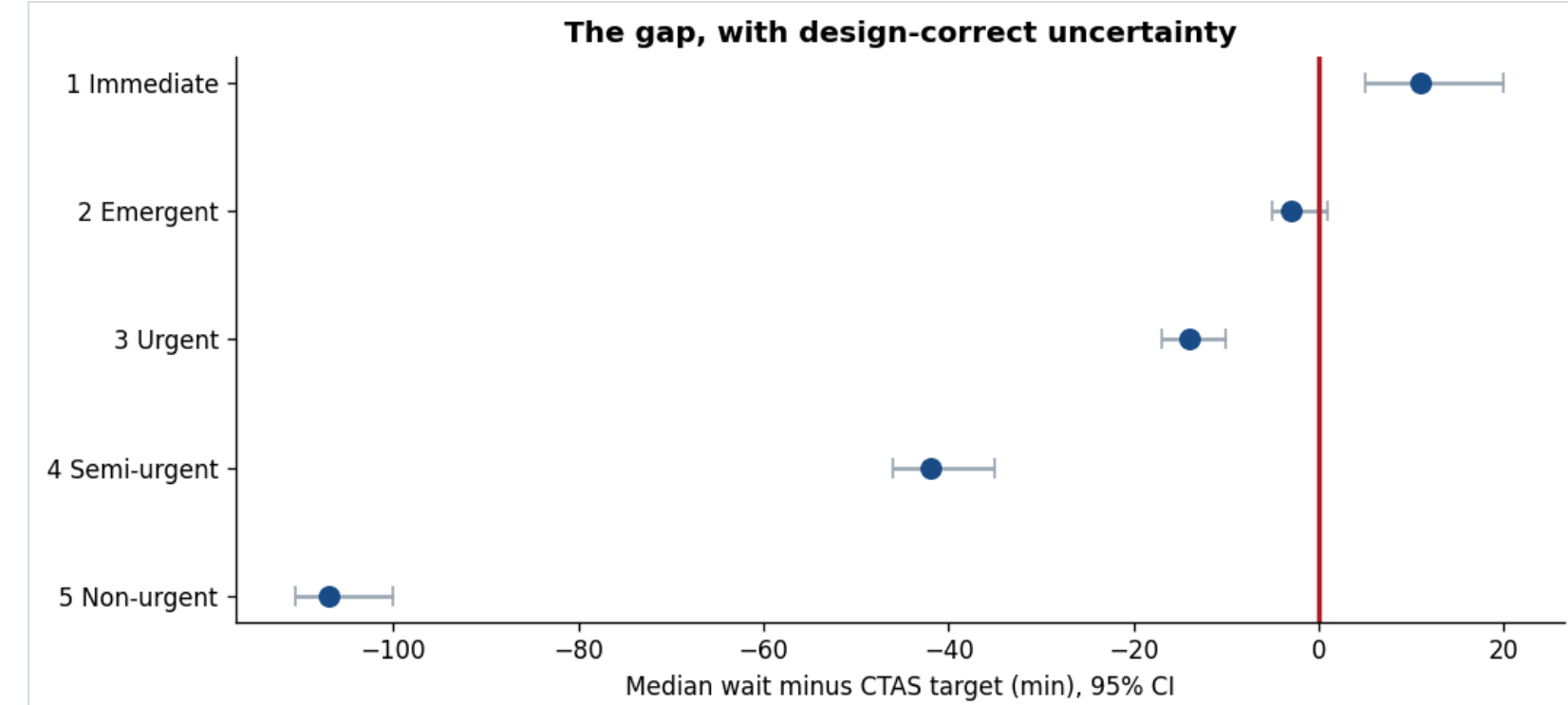
Everyone waits about the same amount of time

Median time to a provider is **11, 12, 16, 18 and 13 minutes** across levels 1 to 5. That is nearly flat. The targets those waits are judged against are **0, 15, 30, 60 and 120 minutes**. That is steep.

Put a flat line against a steep line and the gap between them **changes sign**. The sickest patients end up over their target. The least urgent end up far under theirs.



Left: the five wait distributions sit almost on top of each other. Centre: observed wait against clinical target. Right: the gap, which flips from positive to negative.

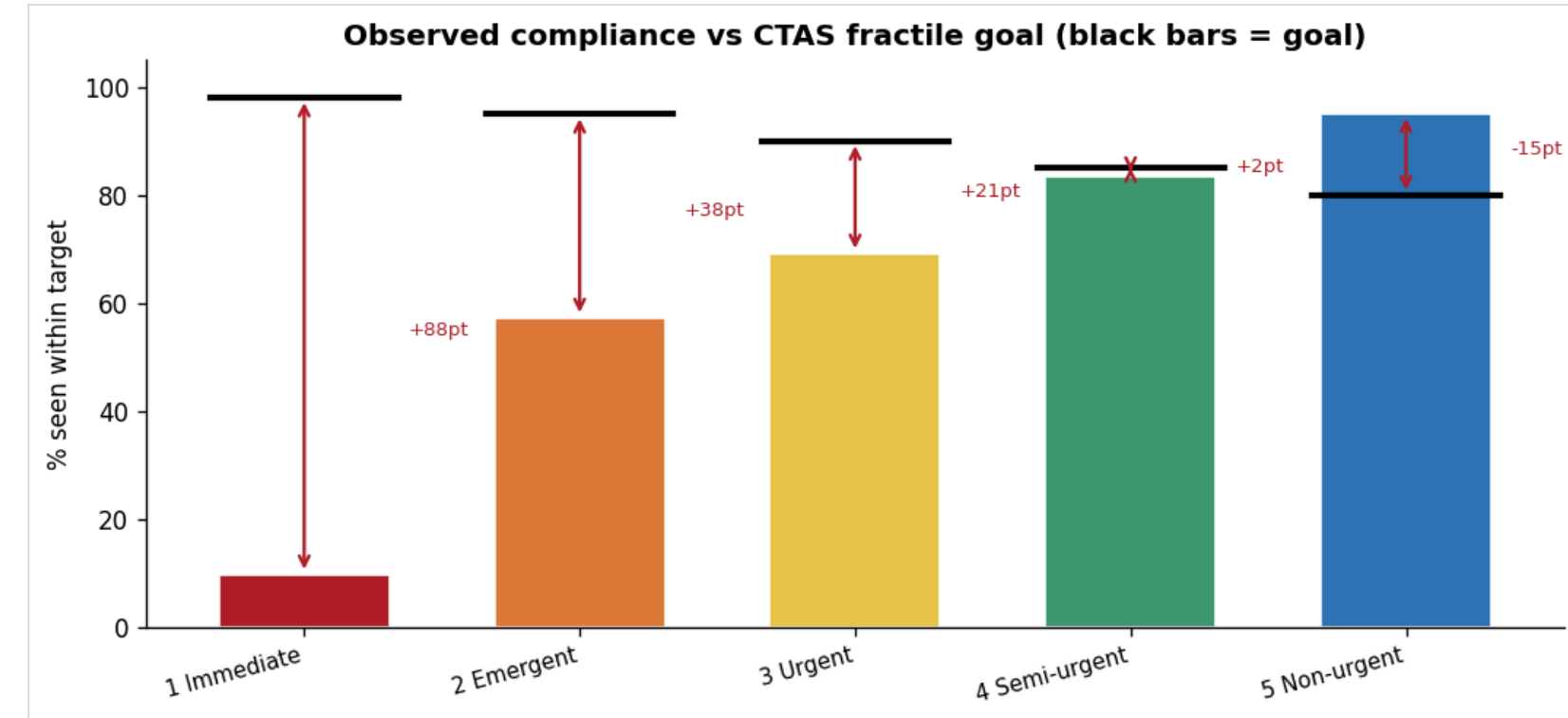


The pattern survives proper statistics. With survey weighting and hospital clustering applied, level 1 sits significantly **over** target and levels 3 to 5 sit significantly **under**. The intervals exclude zero.

Compliance collapses at the top of the scale

Reading the same data as a pass rate makes the problem concrete. The clinical standard sets a target wait **and** a goal for how often you should hit it.

Urgency level	Seen in time	Goal	Shortfall
1 Immediate	5.5%	98%	92 points short
2 Emergent	55%	95%	40 points short
3 Urgent	68%	90%	22 points short
4 Semi-urgent	80%	85%	5 points short
5 Non-urgent	95%	80%	15 points over



Performance is near perfect at the bottom of the scale and near total failure at the top. An ED watching only its average wait time would never see this, because the average hides the level it belongs to.

READ THE TWO COLUMNS TOGETHER

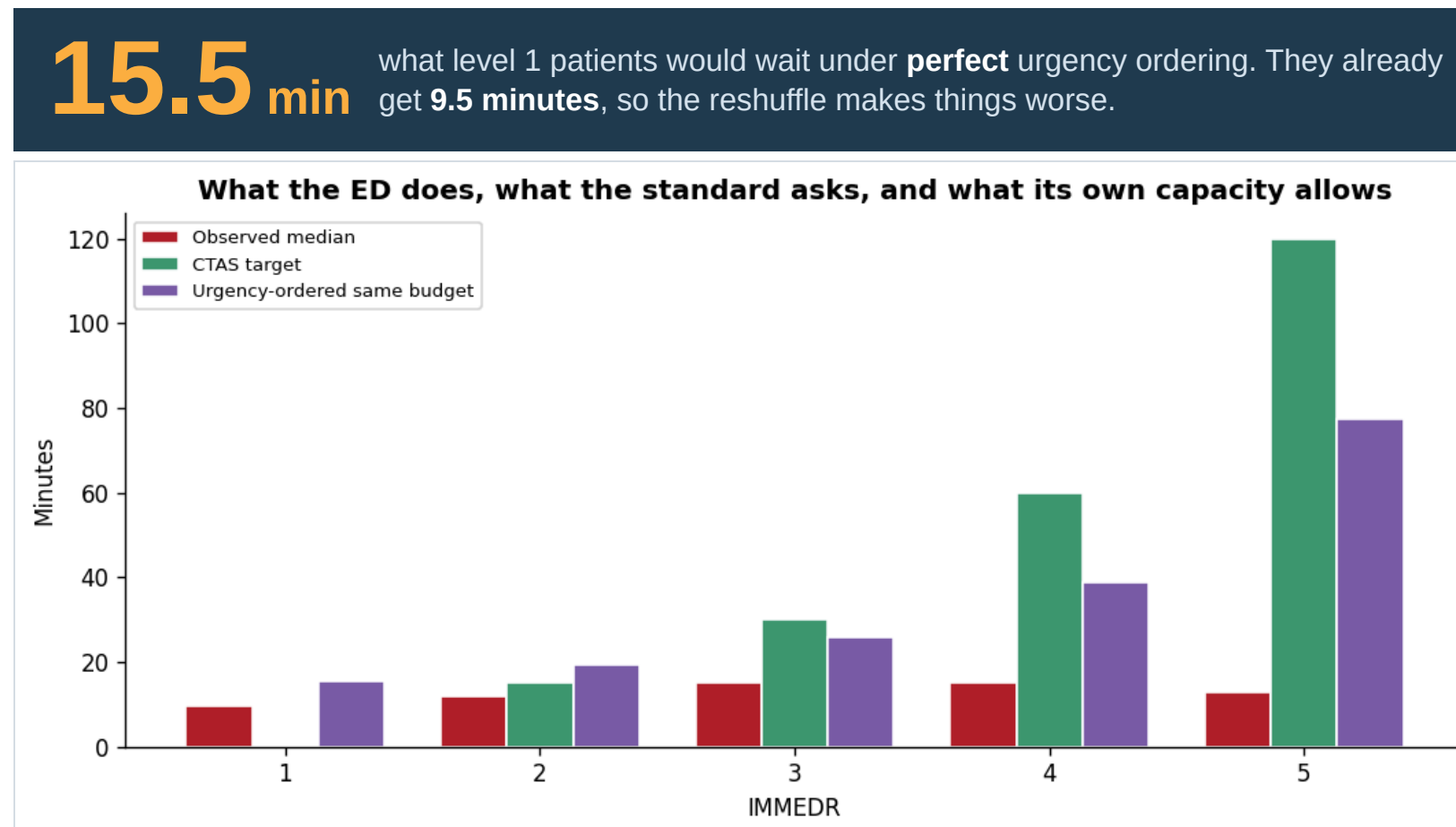
Level 5 clears a goal of 80% with a 95% pass rate. Level 1 misses a goal of 98% with a 5.5% pass rate. Same department, same shift, opposite outcomes. The standard was built assuming wait time tracks urgency, and in practice it does not.

3 WHY IT HAPPENS

Two tests that rule out the obvious explanations

It is capacity, not priorities

The intuitive fix is to tell staff to triage better. We tested that directly instead of assuming it. Holding each ED's **total minutes fixed**, we re-dealt every minute strictly in order of urgency and asked what the sickest patients would get.



Red is what EDs do now. Green is the standard. Purple is a perfectly urgency-ordered day. Even the purple bar misses the level 1 target, which means no reordering of existing capacity can close the gap.

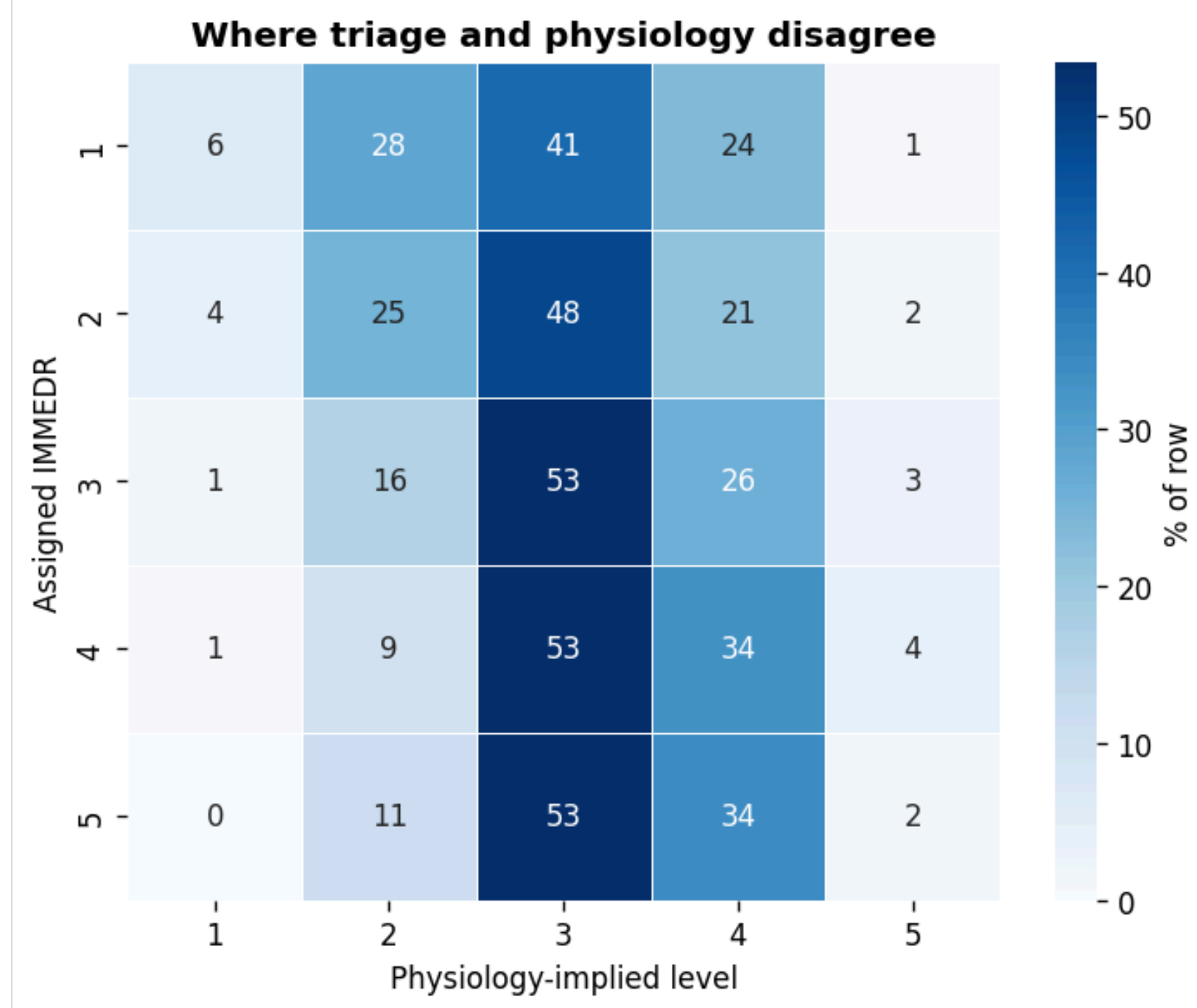
WHY THIS CHANGES THE RECOMMENDATION

These hospitals already prioritise their sickest patients better than strict urgency ordering would. Retraining triage staff will not move the number. Only added throughput will.

Why a triage AI would have failed

We built the "predict urgency from vital signs" baseline anyway, so we could measure exactly how learnable the task is.

$\kappa = 0.16$ agreement between what the **vitals imply** and what the **nurse assigned**. Close to zero. Intake data barely constrains the decision.



Every row collapses toward the middle. Vitals push nearly every patient to level 3 no matter their real urgency. A classifier trained here learns the hospital's habits, not the patient's condition.

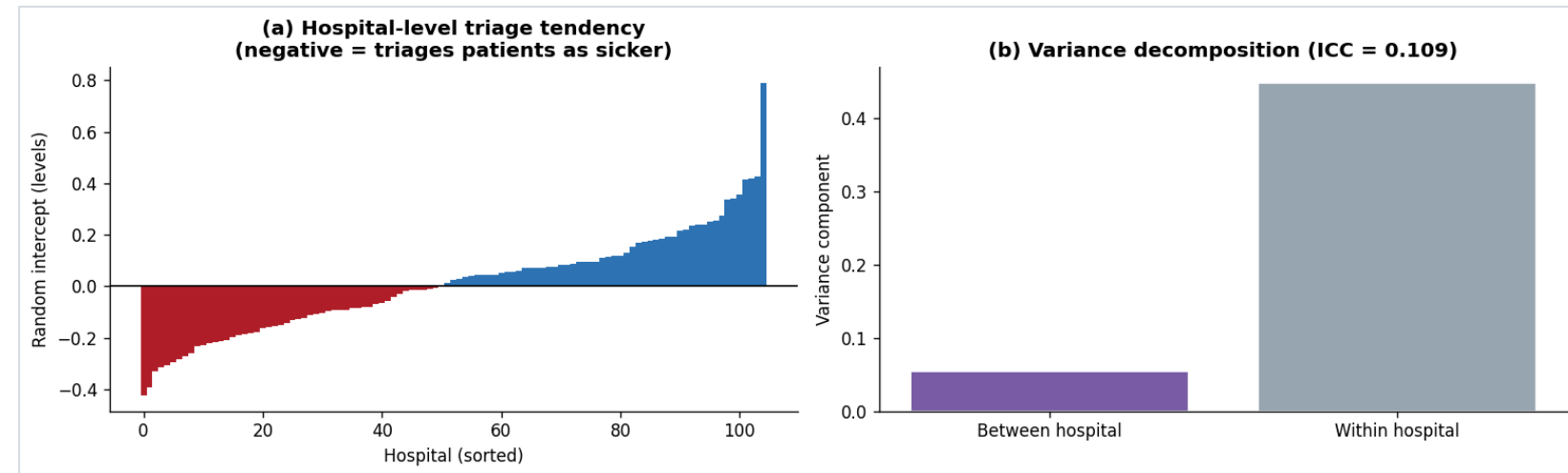
The hardest boundary is **level 2 against level 3** (AUC 0.65), and it carries the most patients. That is where clinicians are working on judgement alone, and where decision support would actually earn its place.

4 WHAT IT MEANS

Structure hospitals can audit, act on, and build from

Your triage score depends on your hospital

After adjusting for how sick patients actually are, **10.9%** of the variation in assigned urgency is explained by which ED a patient walked into.



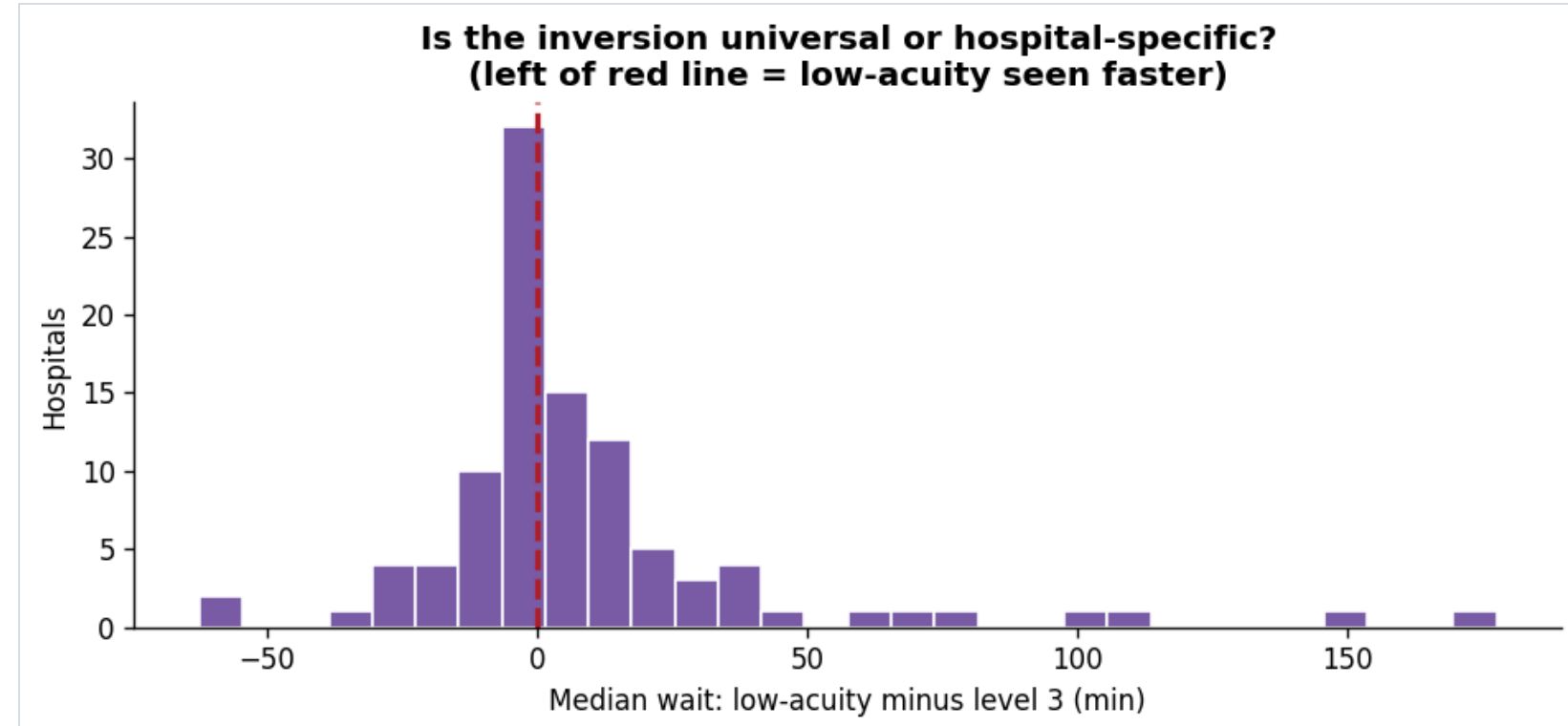
Left: hospitals span nearly a full triage level in how conservatively they score. Right: that between-hospital variance stays after adjusting for physiology, so it is not case mix.

A BENCHMARK NOBODY IS USING

EDs are audited on speed. No one audits them on **triage consistency**. This number makes that measurable.

Low-acuity patients move through a second queue

Level	Sees an attending	Trainee only
1 Immediate	94%	2.7%
3 Urgent	90%	9.4%
4 and 5	77%	22.8%



Nearly 1 in 4 low-acuity patients never sees an attending, and their visits run a third as long (94 vs 292 min). Across 100 EDs, **39%** already seat low-acuity patients faster than urgent ones. The split is bimodal, so some hospitals run a fast track and most do not.

What a hospital can do with this

- Audit level 1 compliance on its own. A single average wait time hides a 92 point failure.
- Put a second look at the 2 against 3 boundary. Weakest signal, highest volume.
- Benchmark triage consistency, not only speed.
- Fund throughput before retraining. The reallocation test shows ordering is already near optimal.

Future work

- Test redistribution inside one hospital. Levels 4 and 5 already clear quickly and are often handled by separate streams, so the open question is whether minutes can be moved safely between levels 1, 2 and 3 without pushing any of them past target. That needs a within-hospital simulation, not a national average.
- Talk to the people who assign acuity. We can measure the output of triage but not the reasoning behind it. Interviews with ED clinicians on how a level is actually chosen would explain the 11% hospital effect, and is currently a limitation of using survey data alone.
- Extend across years to test whether the compliance inversion is stable or worsening.

LIMITATIONS

Cross-sectional and single year, with elevated 2022 sampling error. Levels 1 and 5 are small (effective n of 88 and 168), so their intervals are wide. Findings describe EDs that document triage. No causal claims are made, and the fast-track mechanism is consistent with the evidence rather than directly observed.

METHODS AND REPRODUCIBILITY

Python, pandas, statsmodels, scikit-learn, lifelines. PSU cluster bootstrap (1,000 draws), proportional-odds ordinal model, hospital random-intercept ICC, Jonckheere and Terpstra trend tests, Benjamini and Hochberg FDR at $\alpha = 0.01$. Every cleaning rule and constant sits in one config cell. Full notebook and IEEE paper available on request.